

KEEPING TIME

Atomic clocks, which are used in technologies such as satellite navigation, keep precise time using the properties of certain atoms such as those of cesium-133. When atoms are exposed to photons, some electrons jump between energy levels. When the incident photons have precisely the same frequency as a cesium-133 atom, electrons in the atoms resonate and leap between energy levels. One second is defined as 9,192,631,770 oscillations at that frequency.

Measuring time

The most precise way of measuring time is based on using the frequency of microwave radiation that excites electrons to jump between energy states in certain atoms.

The oscillator fires microwaves set to a specific frequency at the atoms, causing them to jump to a higher energy state.

QUANTUM LEAP

Cesium atoms are ionized and fired through a magnetic gate that filters out any with a high-energy state. The low-energy-state atoms then continue on to the radio wave oscillator.

CESIUM-133
ATOMS FIRED

MAGNET

MAGNET

HIGH-ENERGY-STATE
ATOMS REMOVED

